

Risk Scoring Agent (Agent 12)

Deep Technical Specification & Operational Manual

Executive Overview & Purpose

The Risk Scoring Agent consolidates all downstream agent outputs into a normalized, deterministic risk posture: per-asset risk scores, an overall session risk rating, and a ranked remediation priority queue.

1. Architectural Overview & Workflow

- Position in LangGraph Engine: Penultimate node in Pipelines A, B, A_THEN_B, and D, immediately before ExecReportingAgent.
- Input Consumption: Consumes triage, remediation, threat intel, compliance, cloud, network, and forensics reports.
- State Propagation: Writes state.risk_report with scores, rating, and priority queue.

2. Deterministic Scoring Methodology

- Per-Asset Risk Score: Weighted combination of finding severities, CVSS bands, and exposure context.
- Overall Session Rating: Derived from worst-case asset score and critical-finding count.
- Remediation Priority Queue: Orders remediation actions by risk delta per unit of effort (business impact weighting).

3. Transparency & Reproducibility

- Score Breakdown: Every numeric score includes its contributing factors so analysts can audit the math.
- No ML, No Black Boxes: Scoring is a fully deterministic rule set — reproducible across runs.